

Project Title:

Real-Time Adaptive Lighting System – Interrupt-Driven PWM Control with Bang-Bang vs. Proportional Algorithm Comparison

A Fully Simulated Implementation in Wokwi with Timing Validation

The idea is basically to create a smart lighting system that automatically adjusts LED brightness depending on ambient light using an LDR sensor. The LED brightness will be controlled using PWM, and we'll compare two control algorithms:

- Bang-Bang Control → LED fully ON or OFF depending on threshold
- Proportional Control → brightness changes gradually based on light level

There'll also be a push button that triggers a hardware interrupt to manually override the system for 10 seconds.

The whole thing will be simulated in Wokwi, so we won't need physical hardware.

Main things we'll demonstrate:

- real-time sensor sampling
- PWM generation
- interrupt handling
- timing analysis
- control systems comparison
- low-latency response

We also plan to measure:

- PWM jitter
- interrupt latency
- settling time
- loop timing consistency

Components are simple:

- Arduino Uno
- LDR
- LED
- push button
- 10kΩ resistor
- 220Ω resistor